

The Impact of Code-Switching on Students' Participation and Collaboration in University Settings

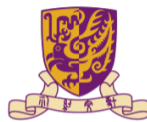
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Abstract

This research studies about the affects of code-switching for university students, especially focused on the traditional Asian university. 99 undergraduate students from all over the world joined this research by answering the questionnaires. The study examines and analysis the relationship between the code-switching frequency and the engagement to the university life through 3 dimensions which are the behaviors in classroom, performance in group discussion and the situation in extracurricular activities. Quantitative results is conducted by using regression analysis, t test and binary regression analysis, which show that code switching in class, extracurricular settings and group discussion is positively associated with higher participation. Differences between local and non-local students are not significant. Qualitative findings show that a useful strategy to improve the enthusiasm of students to join whether the academic stuff or non-academic stuff. It also help the university stuff to deal with the different situations from local and non-local students.

1. Introduction

There are over 7000 languages spoken in the world (Ethnologue, 2023). And this number is continuously growing. Although sometimes English is used as a common language to do the communication. But still a lot of person want to use different one and switch between them instead of using single language. Some scholars like Garcia (2025) thought that is because they want to keep the sense of belongings. Such a sense of belonging exists in both get involved into new environments and the recognition of the personal identity, which is specifically reflected in the willingness in the collaboration and participation in the campus. The staff in the university could assist the students better and that is the reason of importance. Similarly, , in his study of Mainland Chinese students in Hong Kong, showed that code-switching played a complex role in how students negotiated participation and identity in a second language environment.

Existing research has explored this topic a lot and some of them thought that is supportive for students in terms of academic (Yim & Clément, 2021). Yim and Clément (2021) think benefits of code switching during the class is related to students' level. And there are also some scholars have the opposite views, which think the non-native students in the U.S. tend to use their mother tongue to communicate and show a not good performance (X. Li & Jia, 2006). And Elkins and Hanke (2018) considered that Code-switching highlights the privilege of middle-class norms in higher education, as working-class students must learn to 'speak right' and behave according to dominant cultural expectations. There are still some scholars thought that the code switching could do both sides effects. One obvious study said students could benefit from it and suffer from it (Wright et al., 2022). Similarly, Sung (2019), in his study of Mainland Chinese students in Hong Kong, showed that code-switching played a complex role in how students negotiated participation and identity in a second language environment.

Though the existing study have explored the code switching a lot, they lack of the study in the traditional Asian universities, which doesn't have a number of immigrations and keep the single language environment. Moreover, most research only show the abstract psychology instead of the specific behavior.

The overall goal of this study is to study how code switching affect students' behaviors within traditional Asian university. Specifically, it seeks to examine the specific affects like the relationship between engagement and code switching. Due to this study focus on the traditional Asian university, it will fill a critical gap in the literature. Furthermore, the staff worked in the university is going to know more about students and be guided about how to assist the students better for building a more warm environment.

2. Research Questions

1. How does the frequency of code switching relate to students' participation and collaboration?
2. Are there significant differences between local and non-local students in code switching habits and engagement outcomes?

3. Methodology

This study adopted a research design that combined quantitative and qualitative approaches to examine how code-switching influences student participation and collaboration in a traditional Asian university environment.

3.1. Participants

This study targets undergraduate students at CUHK-Shenzhen, with an total of 99 valid questionnaires collected. All students took this questionnaire are from various dialect regions and countries to keep the diversity. The Chinese region is divided because we consider the non-mandarin native speaker, like Cantonese native speaker show a different performance in this situation(Shum et al., 2023). So is other none Mandarin regions. And this study distinguish local and non-local by Mandarin native.

3.2. Data Cleaning

This study is explored through 3 dimensions: Behaviors in classroom, performance in group discussion and the willingness in extracurricular activities.

Since the data collected by Wenjuanxing was not all number-data, they was transformed into a 1-5 grades scale:

$$CS_{Class}, CS_{Extra}, CS_{Group}, Engage_{Class}, Engage_{Extra}, Engage_{Group}$$

and some other special grade to describe. The Mandarin native speaker will be given to 1 and non-Mandarin speaker will be given to 0. The study also made the specific frequency and engagement of 3 dimensions into 1-5 grades scale. The extracurricular part is special because different role in the extracurricular, different weight is distributed to them according to the answer of the role of activities, this weight is in the interval $(0,1)$ ($W_{role-grade}$) and then times the initial grade of

extracurricular dimension:

$$Engage_{Extra} = W_{role-grade} \cdot Engage_{Initial}$$

However, these grades are too discrete to do any data analysis and due the instability when answering the questionnaires. Two steps was done for adjusting this problem.

(1) Smoothing of the 1–5 Likert Scale To introduce continuity while preserving the ordinal structure, a small random perturbation was added to each raw score $x_i \in \{1, 2, 3, 4, 5\}$.

$$\tilde{x}_i = x_i + \varepsilon_i, \quad \varepsilon_i \sim U(-0.35, 0.35)$$

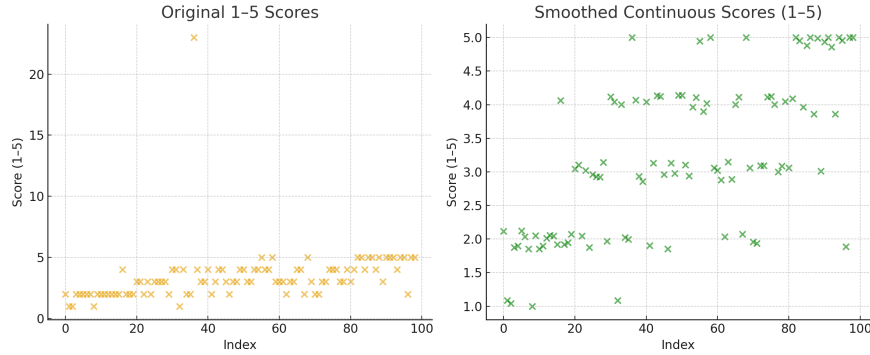


Figure 1: Smoothing Transformation

This smoothing step spreads each discrete point into a narrow interval, reducing overlaps in visualisation and providing a more realistic representation of individual variability, because the perturbation is small, the relative ranking among participants remains unchanged. Tukey (1977, p. 581) and Wickham (2016) both proved this statement and it does works for this study.

(2) Linear Rescaling to a 0–100 Scale To improve interpretability and align the engagement score with common percentage-based scales, the smoothed values $\tilde{x}_i$ were linearly transformed into a 0–100 range using:

$$y_i = 100 \times \frac{\tilde{x}_i - 1}{4}$$

This mapping ensures that:

$$\tilde{x}_i = 1 \Rightarrow y_i = 0, \quad \tilde{x}_i = 5 \Rightarrow y_i = 100,$$

while maintaining proportional differences between intermediate values.

3.3. Mathematical modeling

To avoid zero weights while preserving the relative importance of different contexts in which codeswitching occurs, we apply a smoothed weighting model.

Let the raw context scores be:

$$G = \begin{cases} 3, & \text{if "group discussion" is selected;} \\ 0, & \text{otherwise.} \end{cases}$$

$$C = \begin{cases} 2, & \text{if talking with peers / class is selected;} \\ 0, & \text{otherwise.} \end{cases}$$

$$O = \begin{cases} 1, & \text{if after-class activity is selected;} \\ 0, & \text{otherwise.} \end{cases}$$

To ensure that all weights remain positive, we introduce a smoothing constant

$$\varepsilon = 0.1,$$

and define the normalized weights as:

$$W_g = \frac{G + \varepsilon}{(G + \varepsilon) + (C + \varepsilon) + (O + \varepsilon)},$$

$$W_c = \frac{C + \varepsilon}{(G + \varepsilon) + (C + \varepsilon) + (O + \varepsilon)},$$

$$W_o = \frac{O + \varepsilon}{(G + \varepsilon) + (C + \varepsilon) + (O + \varepsilon)}.$$

Then the overall mathematical model is built:

$$Engage_{total} = w_c \cdot Engage_{Class} + w_e \cdot Engage_{Extra} + w_g \cdot Engage_{Group}$$

3.4. RQ1: Regression

This study used regression model to examine the relationship between below:

$$Engage_{Class} = \alpha_1 + \beta_1 CS_{Class} + \varepsilon_1$$

$$Engage_{Extra} = \alpha_2 + \beta_2 CS_{Extra} + \varepsilon_2$$

$$Engage_{Group} = \alpha_3 + \beta_3 CS_{Group} + \varepsilon_3$$

And then do the multiple regression to test the relationship simultaneously:

$$Engage_{total} = \beta_0 + \beta_1 CS_{Class} + \beta_2 CS_{Extra} + \beta_3 CS_{Group} + \varepsilon$$

3.5. RQ2: t-test and binary regression

To examine whether the relationship between code-switching (CS) and participation differs between Mandarin-native and non-native students, the variable *Local* was used as a binary indicator:

$$Local = \begin{cases} 1, & \text{Mandarin native,} \\ 0, & \text{non-native.} \end{cases}$$

Independent samples *t*-tests were performed to assess differences between local and non-local students.

To formally test moderation, the overall linear regression models were estimated. The model

included CS, language background, and their interaction term:

$$Participation_i = \beta_0 + \beta_1 CS_i + \beta_2 Local_i + \beta_3 (CS_i \times Local_i) + \varepsilon_i.$$

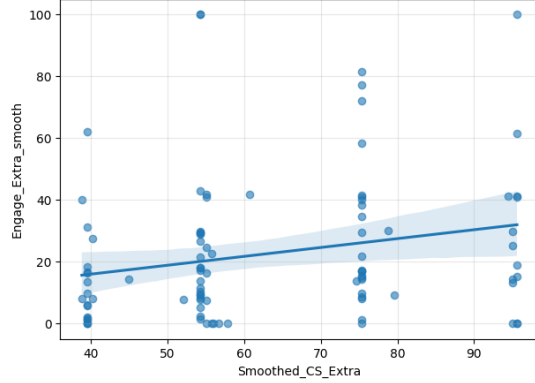
In this specification, the dependent value is adjusted by this binary variable (Fairchild & MacKinnon, 2009). Thus, the key parameter β_3 represents the interaction effect and tests CS–participation relationship. The significance and direction of β_3 address RQ2.

4. Results

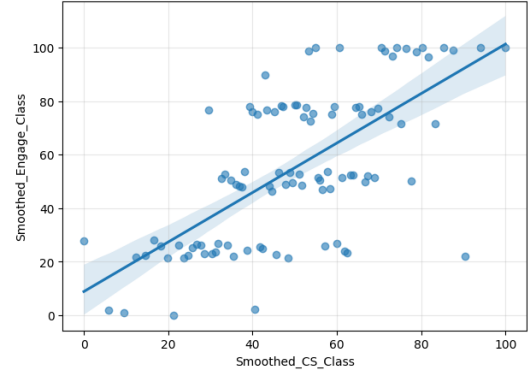
4.1. RQ1 regression results

The scatter map of the three dimensions and the overall dimension is as follow.

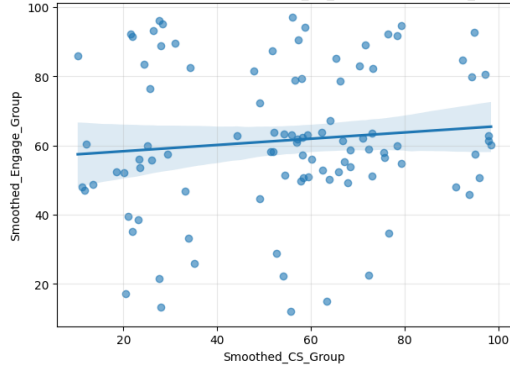
Scatter Plot with Regression Line: Smoothed_CS_Extra → Engage_Extra_smooth



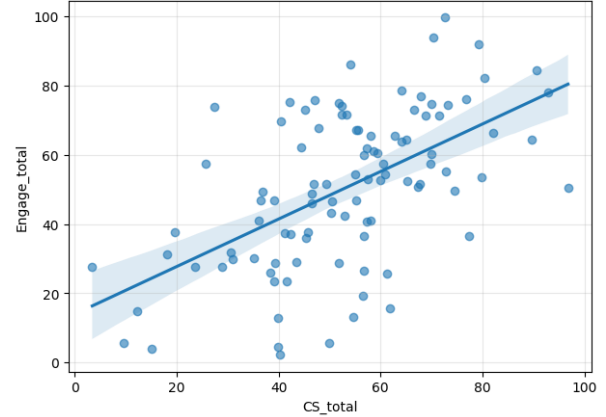
Scatter Plot with Regression Line: Smoothed_CS_Class → Smoothed_Engage_Class



Scatter Plot with Regression Line: Smoothed_CS_Group → Smoothed_Engage_Group



Scatter Plot with Regression Line: CS_total → Engage_total



The class dimensions has the strongest positive relationship much larger than the other two dimensions. The overall regression also showed a strong dimension just according to the scatter map.

Then the precise regression of overall dimension and the multiple regression, the engagement predicted by all dimension variable is shown as follow.

Simple Regression: Engage_{total} predicted by CS_{total}

Predictor	β	SE	t	p
Intercept	13.9954	5.738	2.439	.017
CS _{total}	.6859	.102	6.723	.001
$R^2 = 0.318$ Adj. $R^2 = 0.311$				

Multiple Regression: Engage_{total} predicted by three CS dimensions

Predictor	β	SE	t	p
Intercept	1.1459	6.730	0.170	.865
CS _{Class}	.5990	.099	6.051	.001
CS _{Extra}	.2963	.103	2.882	.005
CS _{Group}	.0331	.074	0.448	.655
$R^2 = 0.451$ Adj. $R^2 = 0.433$				

This showed these two models have a not small R-squared and showed the positive relationship is significant.

4.2. RQ2 t-test and binary regression results

Variable	M_{Local}	SD_{Local}	$M_{NonLocal}$	$SD_{NonLocal}$	t	p	Sig.
CS_Extra	58.58	17.91	69.17	17.88	-2.8396	.0057	Yes
Engage_Extra	17.71	21.86	28.28	22.71	-2.3183	.0229	Yes
CS_Group	50.86	24.56	59.08	23.36	-1.6877	.0950	No
Engage_Group	58.88	21.27	65.14	20.50	-1.4727	.1444	No
Engage_total	48.01	21.46	53.94	23.43	-1.2843	.2027	No
CS_Class	51.57	18.74	47.78	23.35	0.8614	.3918	No
CS_total	51.83	16.17	55.09	21.20	-0.8276	.4106	No
Engage_Class	54.88	26.64	55.39	31.42	-0.0850	.9325	No

Only the CS_Extra and the Enagag_Extra showed a significant difference and others don't show the significant difference.

The binary regression result is like follow.

Model: Engage_total ~ CS_total, Local, CS_total × Local			
Variable	b	t	p
Intercept	12.18	1.49	.140
CS_total	0.76	5.47	.001
Local	5.80	0.50	.619
CS_total × Local	-0.18	-0.87	.388
$R^2 = 0.330$		Adj. $R^2 = 0.309$	

This showed that the local variable doesn't really affect the relationship between engagement and code switching.

4.3. Sociological findings

For research question 1, the study found that all those 3 dimensions existed the positive relationship between code switching and engagement of the university life. And the behavior in the

classroom has the strongest relationship. All 3 dimensions affect the engagement simultaneously and all of them have the positive effects.

For research question 2, the study examined the insignificant difference between local and non-local students in terms of all three dimensions. Only the mean and standard deviation of the code switching and engagement in term of the extracurricular activities. Furthermore, local this identity don't affect the relationship.

5. Discussion

Findings indicate that code switching supports communication and participation in university. Students switch languages strategically to improve understanding, express complex ideas, and navigate mixed-language groups. Code switching also helps reduce communication pressure, promoting smoother interactions. This is obviously from the strong positive relationship between engagement and code switching.

Deeply, the class dimension is the strongest one among three dimensions. This is because class is a very formal and sometimes strict situation and the instructing language of CUHK-Shenzhen is English, CS is strategically used by students to facilitate comprehension and maintain participation in English-medium classrooms (Sung, 2016). Lots of student must even do the language switching to finish the requirement in the class and this is a kind of engagement in class.

The lack of major differences between local and non-local students suggests that CS is widely recognized as a practical communication tool instead of showing the personal identity (D. C. S. Li, 2017) . Moreover, local tend to play with local in Asian universities, this lead that people don't show their identity and social class but own the familiarity with every thing. Another reason to explain this is that the international students who came to CUHK-Shenzhen is not English native speaker either, they show a similar performance to those local students.

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6. Appendix

Table 1: Survey Questionnaire Items

Item	Question (English Only)
Q1	Are you a Mandarin native speaker?
Q2	What is your mother tongue?
Q3	Which type of code-switching do you use?
Q4	What is your grade?
Q5	What is your self-rated English proficiency?
Q6	How often do you switch languages in classroom discussions?
Q7	How often do you switch languages in extracurricular or social activities (e.g., clubs, volunteering, competitions)?
Q8	When do you usually engage in code-switching?
Q9	How often do you speak or share opinions in class each week?
Q10	In the past four weeks, how many extracurricular or social activities have you joined?
Q11	What was your role in the most recent activity?
Q12	How often do you use mixed languages during group discussions or meetings?
Q13	What is your group's main communication mode during teamwork?
Q14	Was your most recent group task completed on time?
Q15	How would you rate the overall communication efficiency in your group?
Q16	Do you think code-switching helps you participate more actively in campus social life?